Urban Analytics/ Design

PORTFOLIO

Soroush Saffarzadeh
MSc of Urbanism



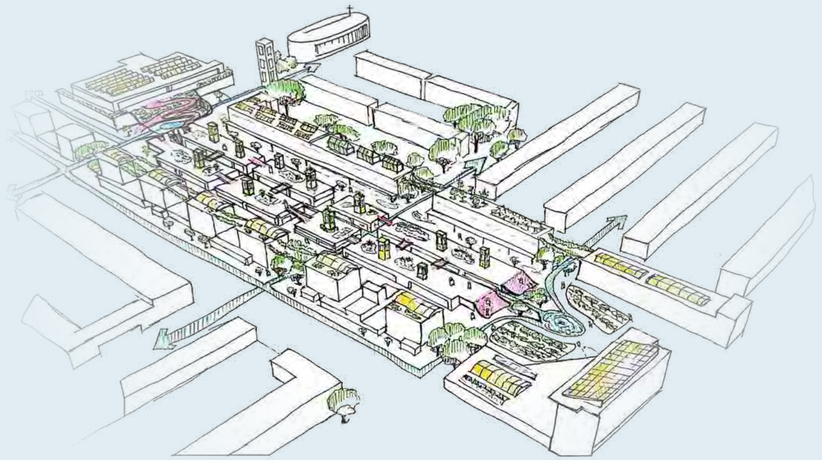
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PARDIS PLANNING

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The Spine of Leyweg

MSc of Urbanism
Urban Environment
Design Project
2024 - 2025



Socio-ecological Design

Q2 Research & Design Studio

The Leyweg shopping center is not just a free-stand shopping building. The shopping area behind the building is a market space with a high potential of becoming a vibrant and livable public space.
There can be a space open to everyone as a ground for social interactions and other activities beside ordinary shopping, where the memories are shaped and promoted. The vitality, vibrancy, and sense of place within a sustainable neighborhood with passive energy design is feasible along with the creation of triggering elements, redesigning the market spaces and the gates to Leyweg's identity.
The project went from a master plan to neighborhood scale detailings.

LOCATION / The Hague, Leyweg local market, NL

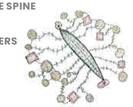
Design approach

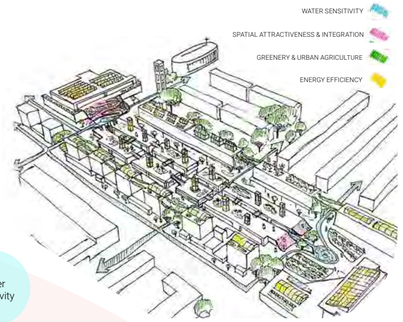
Neighborhood Scale / Climate-adaptive Design



LEYWEG, A SINGLE-USE SHOPPING AREA WITH THE LACK OF ATRACTIVENESS

THE SPINE & CORDS CONCEPT MAKES A SINGLE MIDDLE SPINE AND FEEDS THE CONNECTED CLUSTERS





ILLUSTRATING THE MAIN CONCEPTUAL OBJECTIVES

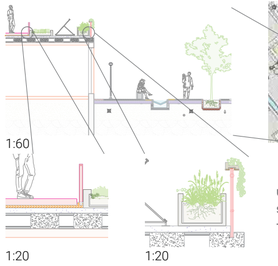
ANALYSIS AND SYNTHESIS OF SCENARIOS











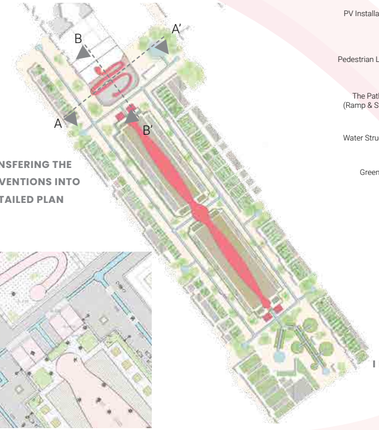


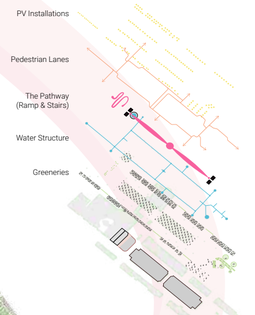
UTILIZING THE GARAGE SPACE AND CREATING THE PATHWAY FORM THE HIGHER TO THE LOWER GROUND

BEFORE AND AFTER CONDITIONS BASED ON THE PUBLIC SPACE DESIGN

TRANSFERING THE INTERVENTIONS INTO DETAILED PLAN







URBAN DESIGN INTERVENTIONS THROUGH THE MASTERPLAN

BEFORE & AFTERS

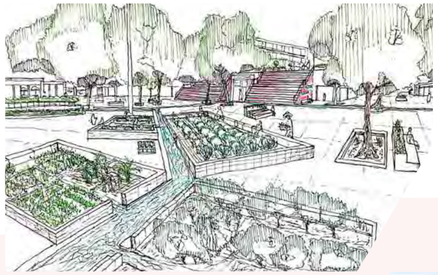




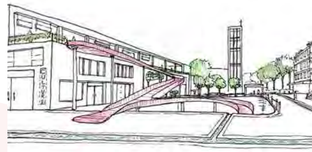


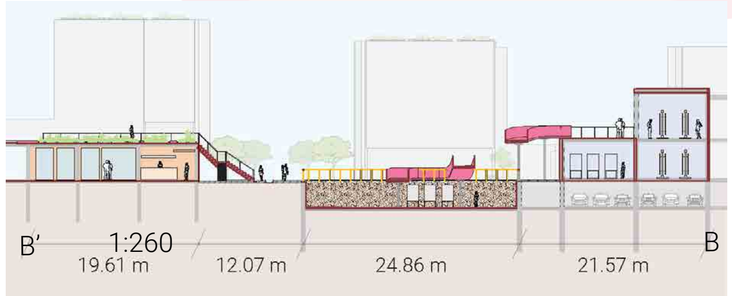
















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Urban Digital Memory

Leveraging Social Media Data on Public Spaces to Inform Urban Design Processes



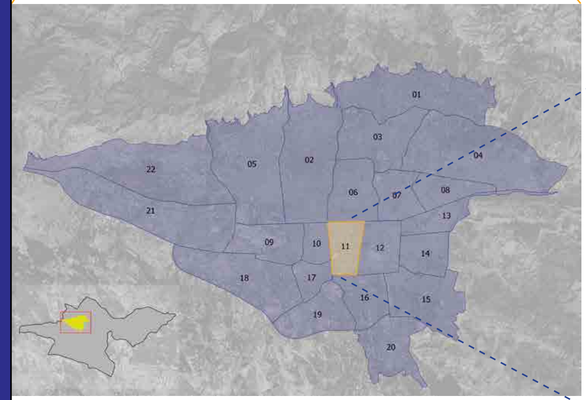


MSc Urbanism Graduation Thesis
Tehran, Iran
2025 - 2026

This study seeks to examine the capabilities of citizens' lived experiences shared on social media platforms as digital memories to value the spatial features of public squares in Tehran. The evaluation is based on emotions expressed by social media users about the public space, whether they have positive, negative, neutral, or mixed impacts. The data will be collected as text strings from X (formerly Twitter) using the Apify scraping app. The collected text data went through topic modeling and sentiment analysis to extract emotions in the form of negative, positive and neutral values. The classified emotions were associated with the spatial features addressed by tweet descriptions through a cognitive approach to spatial perception and the expert's interpretation. The valued associations shaped restrictive, reconsidered, or no-action mitigations. At the end, a design framework was proposed and applied to the case study. The project brought the concept of right to the city, not only as a physical or mental entity, but also as a digital memory, based on the fact that public space is validated by people in digital spaces, which grants them the right to the memory as well. The other aspect of the study would be the passive public information on design provided through digital inputs, giving an initial overview to designers, planners, and decision makers of an urban space.



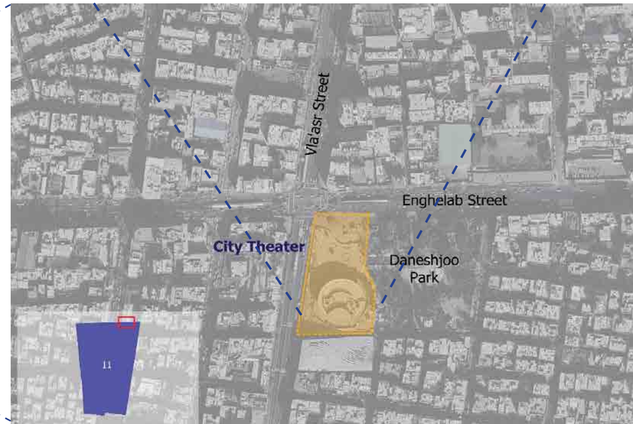
Tehran Province in Iran and Tehran's city divisions



City Theater

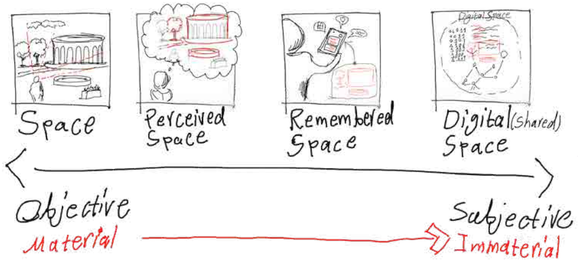
The City Theater Square holds one of the oldest theaters in Tehran, dating back to the 1970s. The City Theater of Tehran building was built as an art and cultural complex. City Theater was not only a venue but also a taste of contemporary design inspired by Iranian architecture (Mohit, 2026; wikipedia, 2026). The layout, presence of the building and its surroundings, formed a public space, as the complex was hosting cultural, artistic and social activities within.





Problem Statement

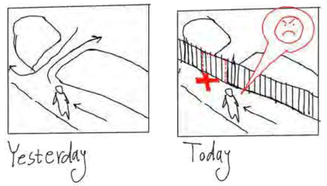
Spatial memories are valuable insights into a space that are taken for granted by designers/decision-makers in their intervention processes, leading to the erasure of public space.



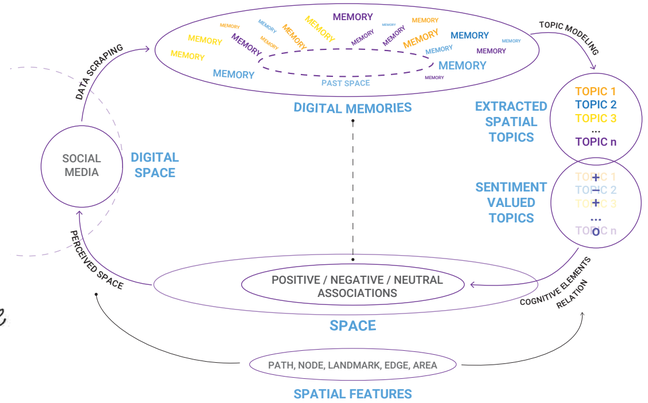
Sense of Place, Digital Memory, Sentiments, Citizen Involvement

Main Research Question

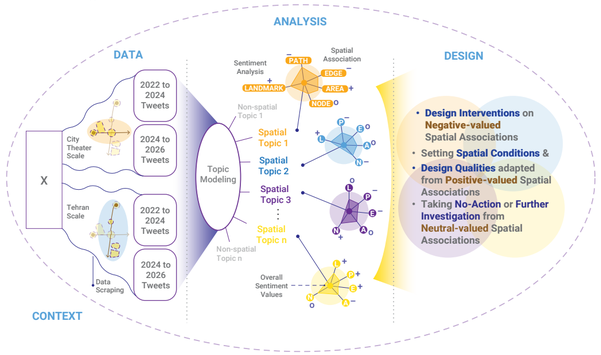
How can digital memories shared by citizens help identify spatial values as potential conditions for interventions in Tehran's public spaces and the City Theater square?



Conceptual Framework

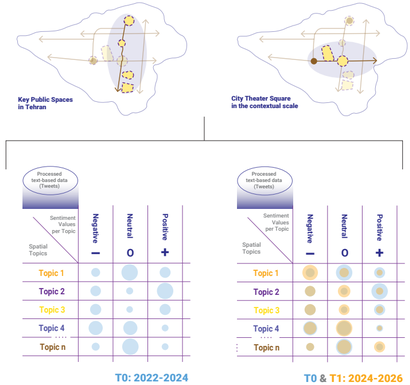


Methodological Framework





The project focuses on how the right to the memory can serve as a significant guiding value for urban interventions in spatial features through a digital citizen-informed data perspective.



Wordcloud of spatial topics & top terms driven from Tweets between '22-'26



Chosen Topics & Representative Tweets

Tehran/Valiasr Street Life & Nostalgia

"How tight I miss the Valiasr and Vanak"

City Theater Metro & Underpass

"I lost myself in the city theater underpass again!"

Grand Bazaar & Market Shopping

"Ta'zyieh-khani (Passion Play) of imam Hossein is a ceremonial art of Islam. Tehran Grand Bazaar, Sabzeh Meydan."

Performing Arts & Festivals (CT)

"City theater was the most peaceful space for my tired soul and body in the midst of Tehran's busiest place[...]"

Public Monuments

"Installing Arash's statues is really one of the only nice things they've done in Vanak."

Tajrish-Valiasr Walks

"Can't wait for April 27th and go to Tajrish, and then let me go down so far that I don't feel my feet anymore."

Valiasr Street Connectivity & Length

"I would like to walk from the Mellat Park to Vanak."

Valiasr-Tajrish Tree Preservation & Urban Ecology

"179 trees are dried out and will be cut down in Vali'asr."

Seasonal Walks (Autumn in Valiasr)

"For everyone it is mandatory to come to Valiasr street in Tehran in the autumn, it will revive you."

Emotional Attachment to Tajrish

"That's it! Now that I'm alone, gotta walk to Tajrish and have a delicious ice cream. Then walk a little bit to make it to a café in Ferdows garden. I spend the winter with these [song]."

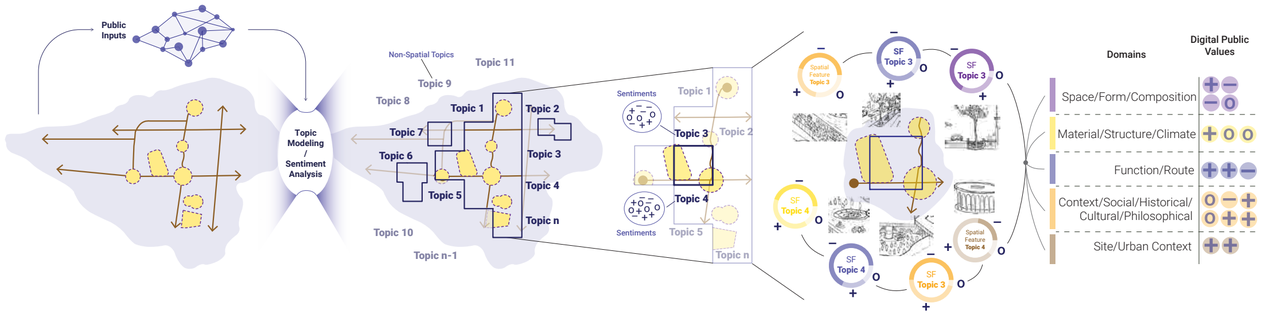
Metro Station Entrances

"Metro City theater at 7pm = the decline of humanity"

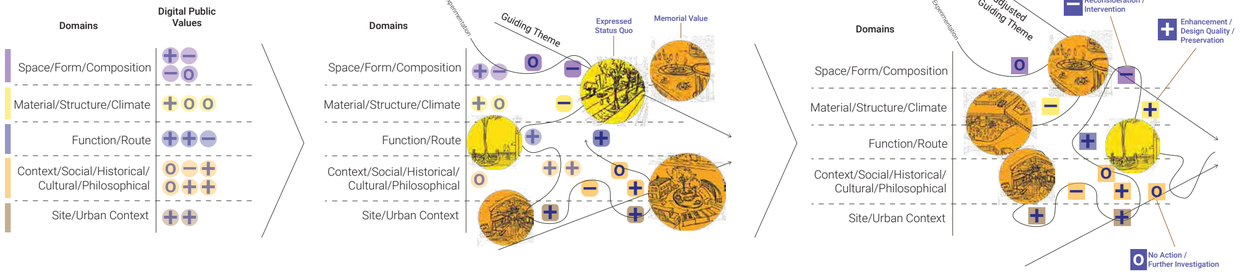
Urban Management & Heritage Protection (City Theater)

"Tehran City theater fountain was destroyed by the municipality."

Design Framework: Linking (A)nalaysis to (D)esign



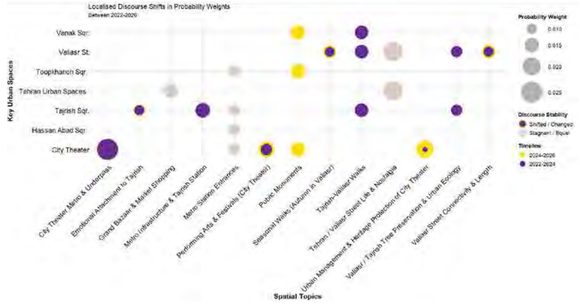
A1. Urban Context | **A2. Accuracy gain with general spatial topics** | **A3. Precision on places with localised spatial topics** | **A4. Quality gain of spatial features with sentiment values** | **A5. Value allocation to domains**



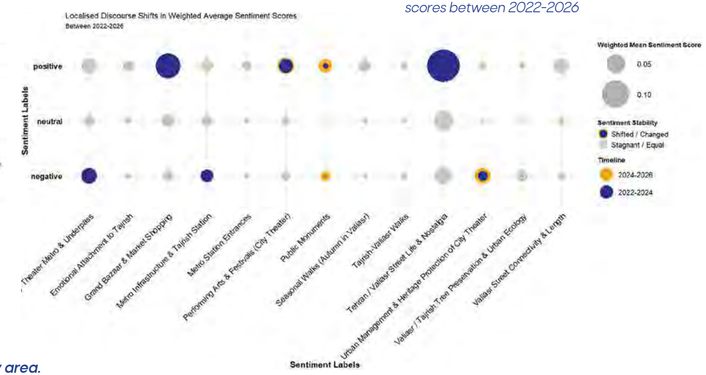
D1. Beginning of Design * | **D2. Design Under Progress With Guiding Values** | **D3. Final Design Process With Applied Guides**

** The design process within the framework is derived from the design process defined by **Van Dooren et al. (2014)** as the literature.*

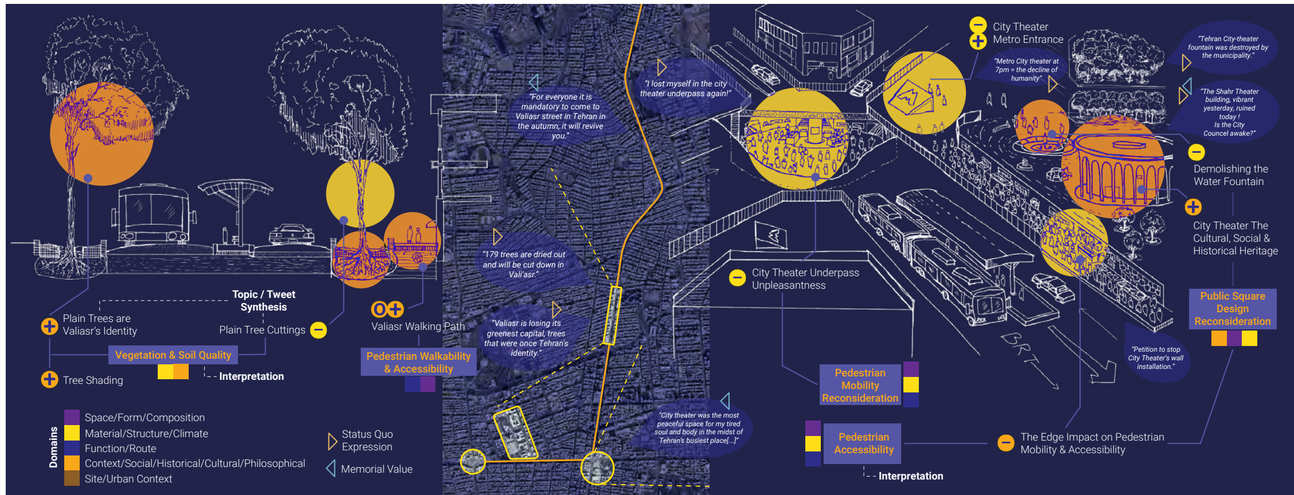
Discourse shifts with probability weight scores between 2022-2026



Topic-sentiment shifts with weighted scores between 2022-2026



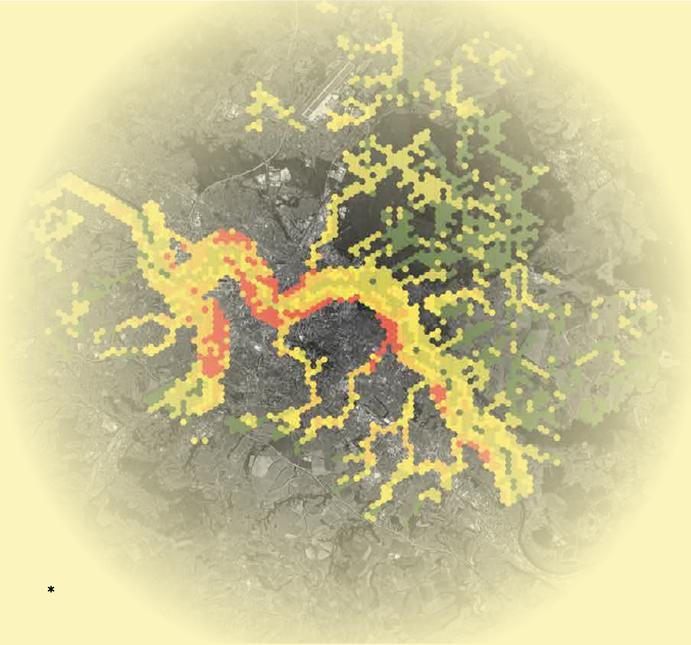
Example of how the (A)nalysis to (D)esign transition was implemented on the study area.



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Unpacking Dresden

MSc of Urbanism
Applied Spatial Analysis Course
June 2025



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Group Project*

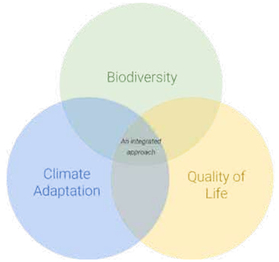
Adriano Mancini
Alankrita Sharma
Alexandre Bry
Grase Stephanie Stuka
Soroush Saffarzadeh

This project was focused on applying spatial analysis utilizing the S-MCDA & Typology Construction as quantitative research methods to evaluate the urban streams within the city of Dresden based on the urban stream restoration goal. The process was started by initial studies of Dresden and its natural elements creating the man-made and ecological synergies within the built and natural environments.

Dresden, Germany

The non-personal contributions are marked with a '' symbol.

Urban Stream Restoration Principles

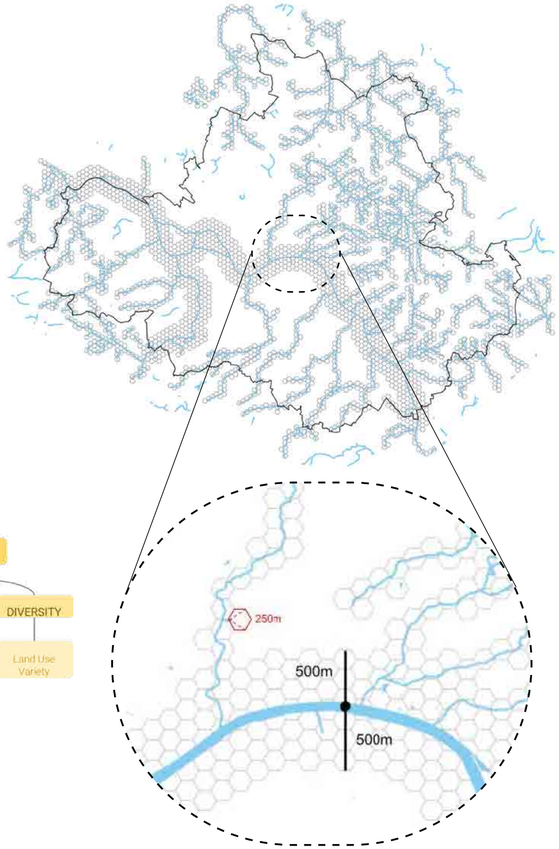


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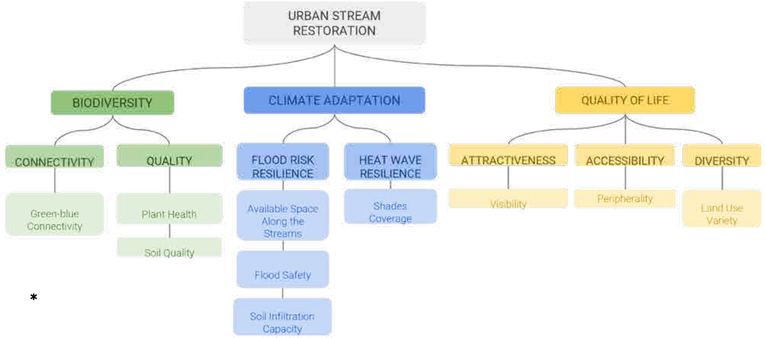
Tackled sub-questions within each domain



Setting the spatial units on the streams & the river



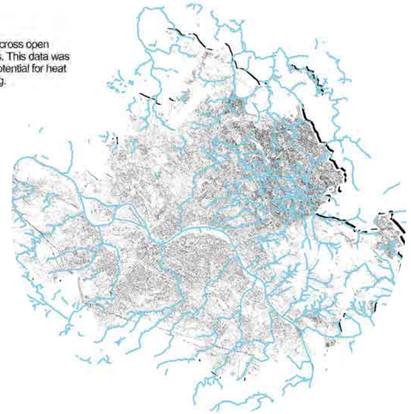
Developing the measurable sub-themes as indexes for the S-MCDA



*

Shade

This map shows shade coverage across open spaces in the stream area networks. This data was used to identify areas with higher potential for heat resilience based on existing shading.



Objective 3: Quality of Life

Shadow Analysis

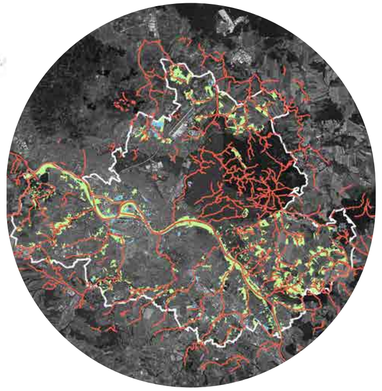
no shade
shaded areas

0 2.5 5 km

1:150,000

Space Along Streams

This map shows available open space areas in close proximity of the streams and rivers of Dresden, useful in understanding potential opportunities for design interventions to prevent, or at least reduce, the flood risk along the streams' banks.



Objective 2: Climate Adaptation

Available spaces along streams
Grasslands
Undeveloped Land

google satellite

0 2.5 5 km

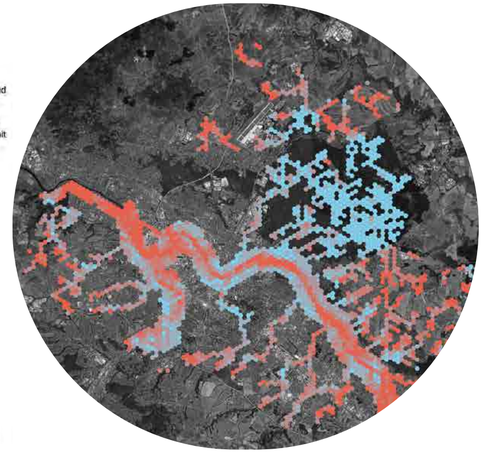
1:150,000

Shade

The DSM and DTM raster layers were needed to run the shadow analysis based on the building and vegetation heights. The output layer with which was a raster layer, was then aggregated to the spatial units. The output values were the median (average) values contained within each spatial unit and were already distributed between zero (full shadow) and one (no shadow) values.

Objective 2 - Climate Adaptation

MCDA Score - Shade at 15:00
0 - 0,083
0,083 - 0,186
0,186 - 0,304
0,304 - 0,451
0,451 - 1



0 2.5 5 km

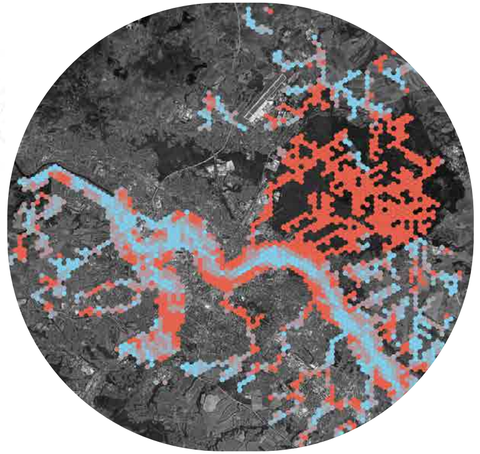
1:150,000

Space Along Streams

The undeveloped land and grassland values layer using the land use ground layer collected from the \[surfdrive\] datasetwas intersected with the spatial units to calculate the coverage space of each land type within each spatial unit before being joined by attributes to the raw spatial units again. This map shows the final S-MCDA scores for availability of space along the streams.

Objective 2 - Climate Adaptation

MCDA Score - Space along the Streams
0 - 0
0 - 0,0074
0,0074 - 0,1706
0,1706 - 0,4296
0,4296 - 1



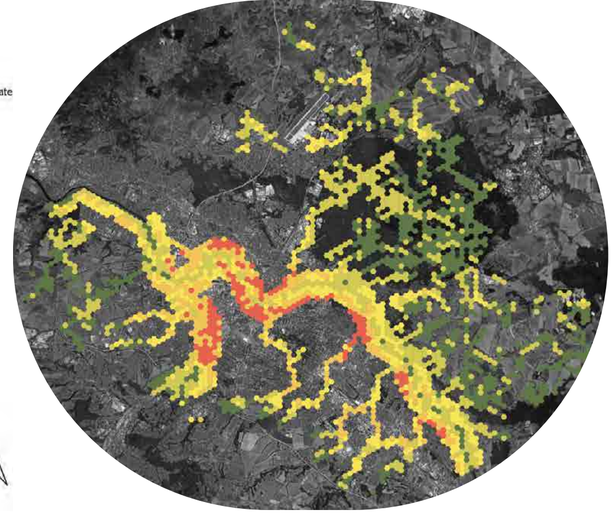
0 2.5 5 km

1:150,000

Combined S-MCDA

The map shows the weighted average scores across the three categories of Biodiversity, Climate adaptation and Quality of Life.

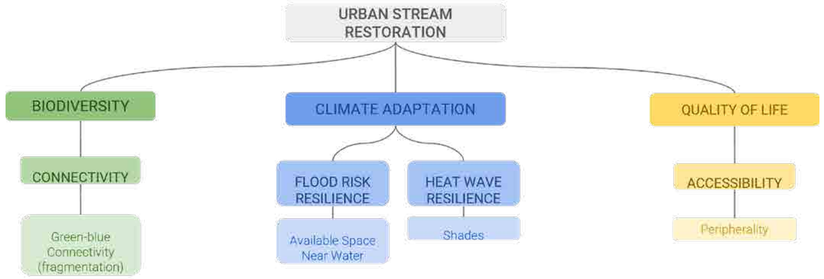
Combined S-MCDA Score
0.197 - 0.2
0.2 - 0.3
0.3 - 0.4
0.4 - 0.5
0.5 - 0.6
0.6 - 0.7
0.7 - 0.7682



0 2.5 5 km

1:150,000

Chosen attributes from the main themes for the Clusters Construction



In continuation of the analysis, three questions were to be answered:

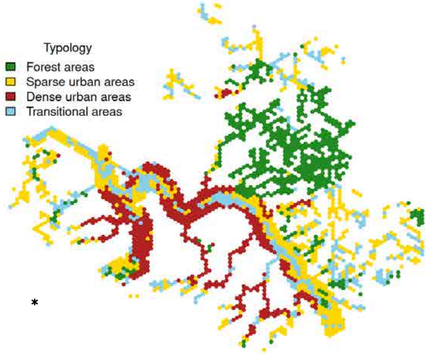
1. What are the streams' adaptability rates to heat waves?
2. How are the accessibility and connectivity conditions of green-blue areas?
3. What are the areas of intervention along the streams?

To answer the questions, the following attributes used for MCDA that were mostly relevant were employed for the typology construction.

- Green-blue connectivity
- Available space near streams
- Shade
- Peripherality

The chosen attributes were utilized in clusters computation which ended up in having 4 clusters containing spatial units that share the most similar range of values and characteristics.

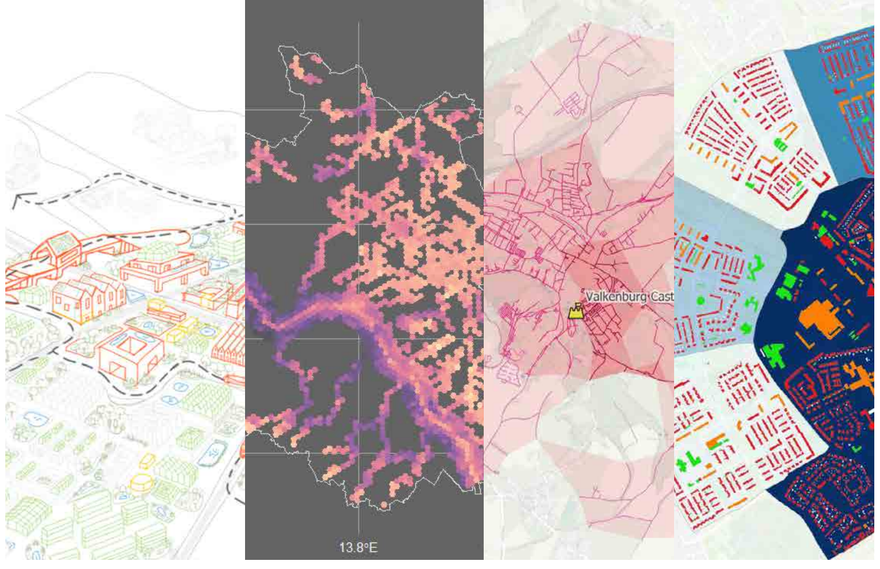
Spatial units clustered with their typology



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Contributions

MSc of Urbanism
Set of drawings and mappings as contributions to group or individual projects
2025 - 2026



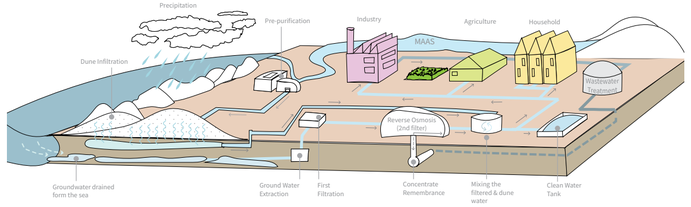
Group Project*
Soroush Saffarzadeh
Youjin Lee
Yaying Hao

The productions that contributed to group projects, mapping challenges or project assignments that can be considered as small independent projects representing my personal experimentations through spatial design and analysis within courses such as Integrated Metabolic Design, #30DaysMapChallenge and GIS course.

Netherlands

*The personal contributions are marked with a ' * ' symbol.

The systemic section of water supply and distributions of companies within Zuid-Holland, such as Dunea. The strategies were developed based on the pattern language approach and implemented on Zotermeer area as the case study.



Dunea water collection, processing and distribution network within Zuid-Holland.



Zotermeer area, at the intersection of urban, agriculture and industry

Strategies based on a collection of patterns focused on sectoral circularity

Fit-for-Use Water Supply

Classify Water Qualities
Seeking extra resources
Handling the valuable water

Make Every Drop Count

Reconsidering activities
Reusable water distinction
Reusable water collection & redirection

Housing-Greenhouse Water Loop

Inter-section water flow efficiency
Water quality maintenance at each reuse step
Water exchange beneficiary

Water Transition with Livable Futures

Urban fabric rearrangement for new water flows
New infrastructure integration
Life quality increase with water transition



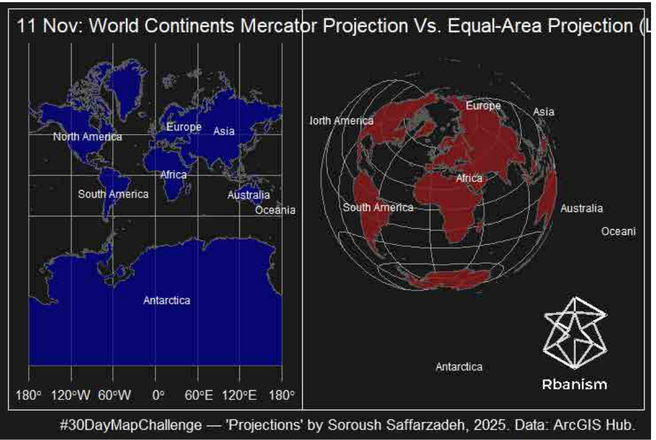
#30DayMapChallenge

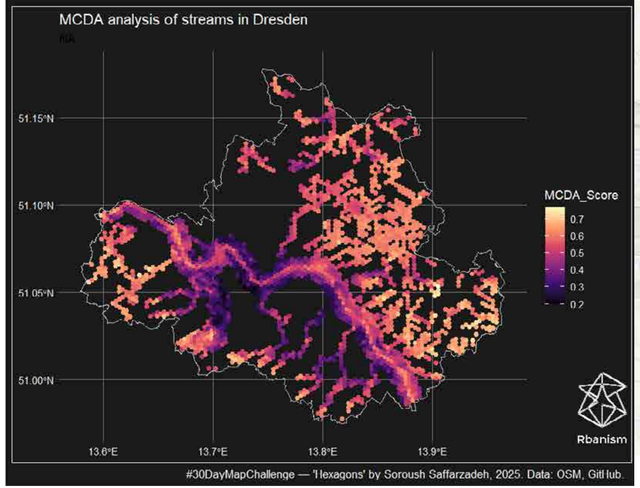
This is a mapping challenge that takes places every November in which you have to produce maps per day based on a specific theme and type of data defined by the #30DMC group. I've produced the maps with specific themes using the R scripts within RStudio environment.

```r
p1 <- ggplot(af_eu_as_merc) +
  geom_sf(size = 0.3, fill = "#ffffff", alpha = 0.7) +
  geom_text(data = af_eu_as_lines,
            aes(x = x, y = y,label = CONTINENT),
            size = 2.3, fontface = "plain", color = "white") +
  theme(panel.background = element_rect(fill = "grey10"),
        plot.background  = element_rect(fill = "grey10"),
        panel.grid.major  = element_line(color = "grey50"),
        panel.grid.minor  = element_line(color = "grey70"),
        axis.text = element_text(color = "white"),
        axis.title = element_blank(),
        plot.title = element_text(color = "white")) +
  ggtitle("11 Nov: World Continents Mercator Projection Vs. Equal-Area Projection (LAEA)") +
  theme(theme(plot.title = element_text(hjust = 0.5)))

p2 <- ggplot(af_eu_as_laea) +
  geom_sf(size = 0.25, fill = "#ffffff",alpha = 0.7) +
  geom_text(data = af_eu_as_lines,
            aes(x = x, y = y,label = CONTINENT),
            size = 2.3, fontface = "plain", color = "white") +
  theme(panel.background = element_rect(fill = "grey10"),
    plot.background  = element_rect(fill = "grey10"),
    panel.grid.major  = element_line(color = "grey50"),
    panel.grid.minor  = element_line(color = "grey70"),
    axis.text = element_text(color = "white"),
    axis.title = element_blank(),
    plot.title = element_text(color = "white"))
```

Mercator and Equal-Area projection comparisons on the world map based on the theme of projections, using RStudio for geospatial data classification, projection and visualization.





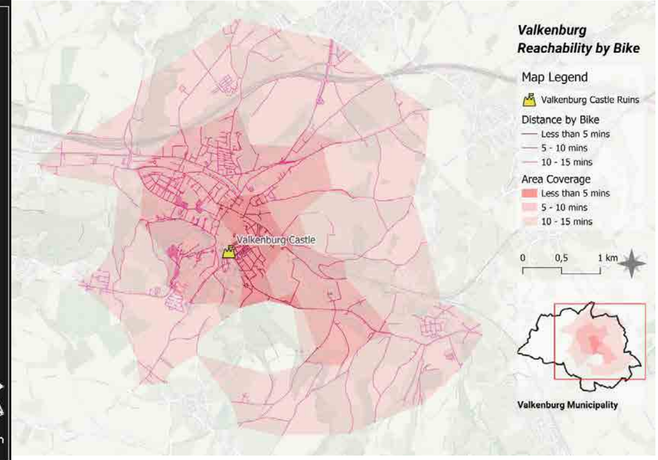

```r
# identifying and locating the values of the layer
head(boundary)
unique(boundary$name)

#filtering the boundary based on Dresden value among other attributes
dresden_b <- boundary %>%
  filter(name == "Dresden")

#load logo
rbanism_logo <- image_read("Logo_Rbanism_ White.png")

# plotting the map
ggplot(data = hex) +
  geom_sf(data = dresden_b, fill = NA, color = "grey75",
          size = 1) +
  geom_sf(aes(fill = MCDA_Score, color = MCDA_Score)) +
  scale_fill_viridis_c(option = "magma" , name = "MCDA_Score") +
  scale_color_viridis_c(option = "magma", name = "MCDA_Score") +
  theme(panel.background = element_rect(fill = "grey10"),
        plot.background  = element_rect(fill = "grey10"),
        panel.grid.major  = element_line(color = "gray50"),
        panel.grid.minor  = element_line(color = "gray50"),
        axis.text = element_text(color = "white"),
        axis.title = element_blank(),
        plot.title = element_text(color = "white"),
        legend.background = element_rect(fill = "grey10", color = NA),
        legend.key = element_rect(fill = "grey10", color = NA),
        legend.text = element_text(color = "white"),
        legend.title = element_text(color = "white")) +
  plot_annotation(
    caption = "#30DayMapChallenge — 'Hexagons' by Soroush Saffarzadeh, 2025. Data: OS
    theme = theme(
      plot.caption = element_text(color = "white", hjust = 1, size = 10),
      plot.background = element_rect(fill = "grey10", color = NA)
    )
  ) +
  labs(
    title = "MCDA analysis of streams in Dresden",
    subtitle = NA)
```

Producing the hexagon map based on the overall MCDA scores of Dresden's urban streams fitted within the spatial units as the data.



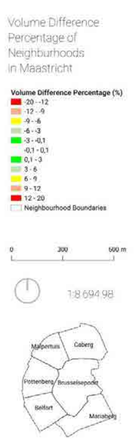
Street network analysis using iso-chrones on reachability coverage by bike to Valkenburg Castle

GIS Operations & Mappings

From taking the course of GIS within the Geomatics programme, I had the chance to advance my knowledge and skillsets using multiple GIS software such as Grass GIS, and FME for complex geospatial operations and analysis on real-world cases.

This led to map productions and visualizations based on the best-fitting cartographic attributes.





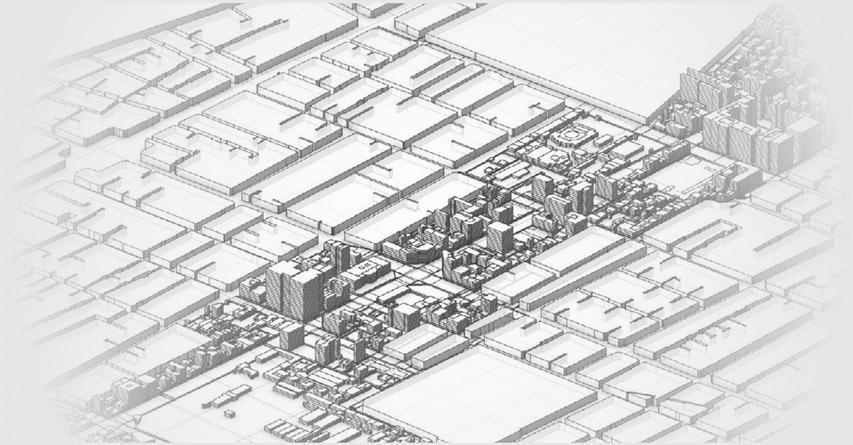
Operations on building volumes & functions their relations to legal standards per municipal divisions in Maastricht.



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Taleqani Urban Form Analysis

Taleqani Urban Design Guidelines Project
2023



Modeling Project

Description

As the kick start of my professional experience in Urban Planning/ Design, I had the chance to work with my university professor, Dr. Mazyar Abaee at Gozineh Consulting Engineers in Taleqani Street Urban Design Guidelines project. I collaborated in the data gathering and analytical outcomes. But one of the prominent works I did with the guidance of Dr. Abaee as the project manager was the 3D modeling of application of current zonings and the visions for the prescribed ones which was performed by CityEngine as the modeling tool.

LOCATION / TEHRAN, IRAN

Design approach

Contextual Study & Analysis



Study Area

Taleqani street was one of the most active and vibrant urban streets in the second Pahlavi era. It was imagined to become the middle east trading and business center, consisting several governmental, institutional and banking offices along side the diverse hotels, retail centers and the embassy of U.S.. After the 1979 revolution in Iran, the street experienced massive changes which led to its current condition. The institutional buildings are intact but other buildings and land uses had to change by their shape or activity.





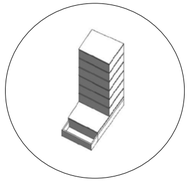
Zoning Codes

At the analysis stage, the data like the floor number or zoning code were gathered from around 1700 parcels (lots) within the study area. After that, the lots were separated to intact parcels and normal parcels. The intact ones were the properties of the government which could not be changed in guidelines or new codings. The normal lots were compared to their prescribed zonings and each zoning code was assigned to them. The whole study area was modeled in two scenarios: the present condition & the coded condition. The coded lots were extruded as same as their zoing codes with the help of the rule-base function of CityEngine software which helped me with extruding the forms based on their zoning codes.

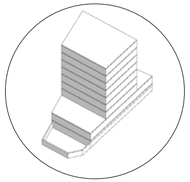




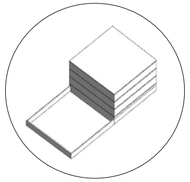
Samples of Prescribed Codings



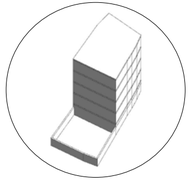
Mixed
Max Flrs: 7
G & 1st Flrs BSC: 80%
Upper Flrs BSC: 40%



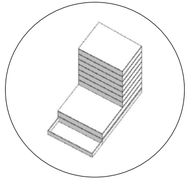
Mixed
Max Flrs: 9
G & 1st Flrs BSC: 80%
Upper Flrs BSC: 40%



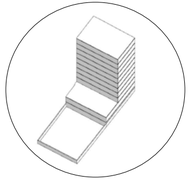
Mixed
Max Flrs: 4
G Flr BSC: 50%
Upper Flrs BSC: 50%



Residential
Max Flrs: 5
BSC: 60%



Service
Max Flrs: 9
G & 1st Flrs BSC: 60%
Upper Flrs BSC: 40%

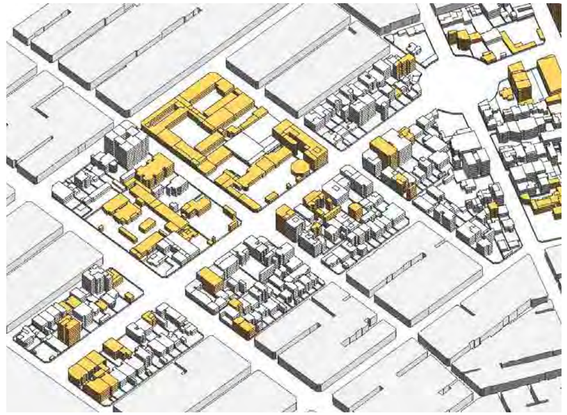


Service
Max Flrs: 11
G & 1st Flrs BSC: 50%
Upper Flrs BSC: 30%

The zoning codes are as shown in figures, describe the activity and formal attributes assigned to each lot such as the land use, maximum floor numbers, first and upper floors . Building Site Coverage (BSC). Each of these attributes are considered in coding rules of lots inside the CityEngine. The final results were modeled and visualised.

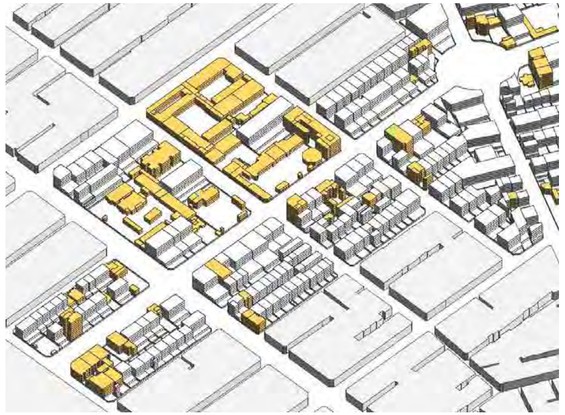
Modeled Scenarios

Current Condition



A part of the study area which is modeled based on its real density and building coverage patterns of the lots beside the intact lots which are colored are shown.

Prescribed Code



In the prescribed code scenario, the normal lots are modeled based upon their new attributes, alongside the intact buildings attached to their sites.

Current Condition



The intact buildings and normal ones had a unique set of diverse building coverage patterns and dimensions which do not follow a common pattern across the fabric.

Prescribed Code



After the coding process, the fabric becomes more ordered and integrated as a bunch of solid forms with different densities.

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Personal Experiments

Experimental Personal Projects
2019 - Present



Curiosity

Just for fun

The following contents are some of my personal works and experiences from my trips and explorations across the cities and urban spaces, Including hand sketches from my sketchbooks. Moreover, there are my experimental attempts with 3D modeling software known as "Rhino" in order to create abstract forms and visualizing them by rendering engines such as Twinmotion to reach a visionary atmosphere at the final stage.

LOCATION / IMAGINATION

Design approach

Experimental

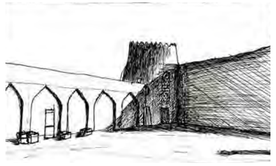
FROM THE SKETCHBOOK









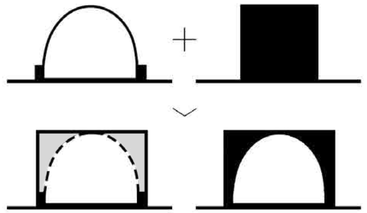





One of my favorite hobbies emerged from a trip in Kashan historical city back in university time. It was the simple sketching using pen or pencil in a small sketchbook. Since then, where ever I found an interesting subject around the corner, I drew it down in my little sketchbook.

THE NEGATIVE DOME

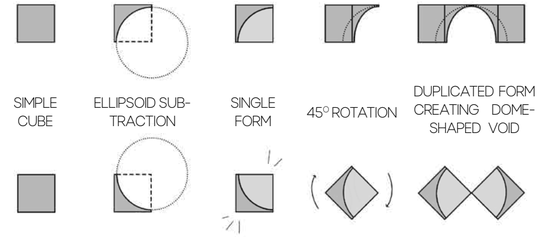
The cube project started back in my second semester university project which was about redesigning a historical square in Isfahan, Iran. There was an abandoned space in the square that made me think of a symbolic object in order to occupy the empty space so that it could order the environment.



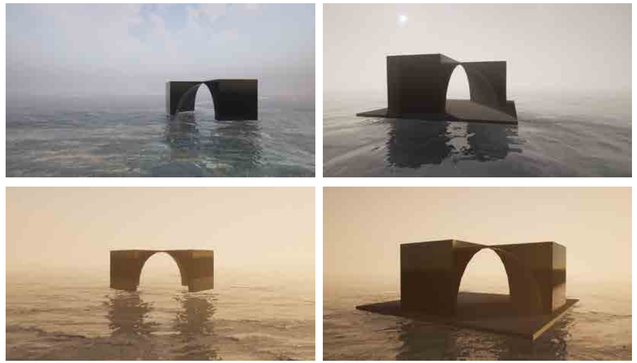
The initial form came from the idea to combine the simple cubic form with the ellipsoid space form which is originated from the dome, a special physical element that is pretty common in Persian historical architecture.
After prototyping a primitive model in Microsoft's 3D printer modeling software, I managed to create the exact forms inside the Rhino environment after watching some basic tutorials about the software and many experimental trials and errors. Good to mention, due to simple UI/UX design of Rhino environment which made the tools usage way too easy, the experiment was pretty fun and joyful.



Front View



Bottom View



/

Pardis Planning

Urban Planning Studio 5 Project
2022



Land Arrangement

Description

As our fifth Urban Planning Studio group project, we were told by our instructors to work on new town of Pardis undeveloped lands. Since Pardis was an under development town, we had the opportunity to commence a land preparation project consisting of a detailed planning plan from the initial site analysis studies, to a final render of planned and designed neighborhood division 3D model. The coursework done by myself are shown in these contents.

LOCATION / NEW TOWN OF PARDIS, IRAN

Planning/Design approach
Sustainable Social Housing

New Town of Pardis

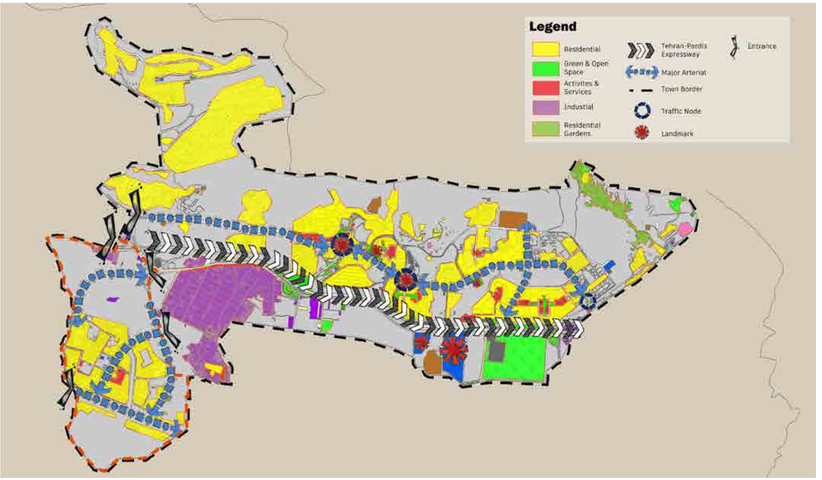
The new town of Pardis is located in the east side of the capital city of Tehran, IRAN. After the mass immigrations to Tehran, around early 80s, the government, municipality and developers decided to take the neighbor lands around Tehran to develop new accommodations for the incoming population so the capital's density and congestion could be controlled. It started with building massive residential blocks on higher grounds and uphills with the help of private developers but due to economical crisis, many of the development plans failed which it left the town with lots of empty spaces and untouched grounds.











Neighborhood Division

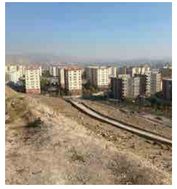


Design Site



The town is divided into 8 districts which the 3rd district was the chosen division. Our project group took the neighborhood subdivision which can be seen in the aerial pictures. Afterwards, we split the neighborhood in two subdivisions and the design site was ready for the analysis.
As the first impression on site visit, we noticed a hilly ground with almost zero development which was surrounded by boulevards and a dirt road. The main challenge was the geographical situation.







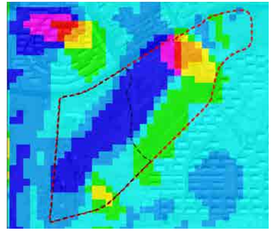


Site Analysis

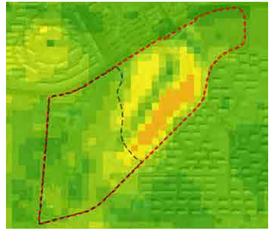
Geographic Analysis & Strategic Views

The initial analysis started with the site geographic analysis in the town. We had to analyse the geographical characteristics. We used the ArcMap analysis tools to assess the slope, aspects, heights and visible spots to find out the planning and development possibilities. As mentioned before, the design site was literally a ground with downward slope. Based on the contour lines, the height spots variations was moderate but consistent. It was literally a vast downhill.

01 Aspect



02 Slope



03 Contour



04 Visibility

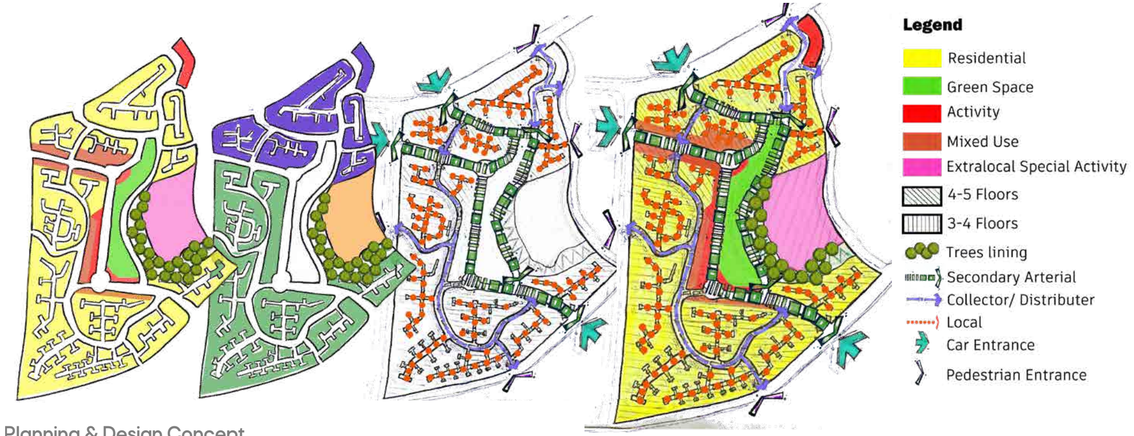






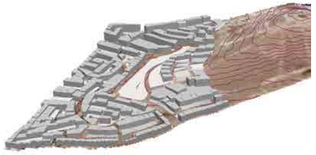
Afterwards, We needed to understand what sort of scapes were available at the present condition, so we used 3D vision on google earth pro to assess the up-down and down-up views of the site. As can be seen in figures, the northern view is a mixture of urban and mountain scape and the southern view is urban scape with roughly-flat lands.

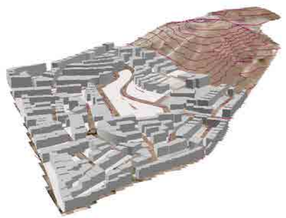
LAND-USE PLANNING | FLOOR DENSITY SETTING 4-5 flrs & 3-4 flrs | NETWORK & ACCESS DESIGN | FINAL INTEGRATION



Planning & Design Concept

Once the site analysis was done, the planning process began with the land use per capita estimation of the predicted population for the next decade of the design site. It helped us to distribute activities and residential areas logically based on the predicted population for a small community.
Although the planning came first, we tried the design concepts at the same time. By hand drawings on design site and contour line patterns, the street networks and nodes had been drawn and blocks were set as an AutoCAD map. There were also a floor density planning set.
The integration of land-use and design concepts, led to a local community with spot attractions on hillside and adequate vibrant spaces based on site characteristics.







3D Modeling

We had to simulate the integrated concept on the real scale and geographic conditions. After assessing the first overall 3d model of the concept design on ArcScene visualizer, we managed to take the georeferenced data layers to a better 3d modeling tool known as the CityEngine. We used the combination of aerial map, topography data, building & lot shapefile layers and street network graphs to reach the optimal vision of our final plan. As the final step, the whole integrated model was sent to 3d visualizing software of Twinmotion for the final presentation.









/
Takhti Neighborhood Design

Urban Planning and Design
Studios II & III
2021-22



Planning & Design

Description

In urban planning studio II, the project groups were told to provide a cognition report about the KhaniAbad neighborhood AKA "Takhti" consisting of the site analysis on all urban planning dimensions including ecology, activity, transportation, architecture and urban form etc. to reach a multi-dimensional and comprehensive knowledge about the neighborhood so it could help us in the urban design guideline stage in urban planning studio III. We used diverse surveying methods and tools for the site analysis and used either hand-drawn design guideline concepts or 3d generated models by SketchUP and CityEngine to reach the final design.

LOCATION / TEHRAN: TAKHTI NEIGHBORHOOD, IRAN

Design approach
Contextual

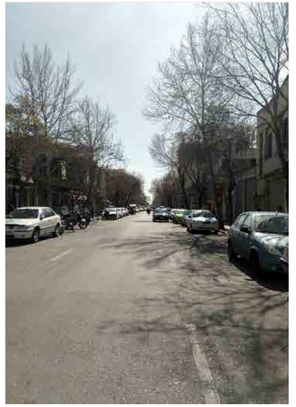
Initial Cognition / TAKHTI

Takhti neighborhood which is known as KhaniAbaad was named after the sports national icon of Iran, Qolam Reza Takhti, which was a popular wrestler who lived in this neighborhood. Takhti was one of the oldest neighborhoods based in district 12 of Tehran city, the core of historical capital, pretty close to the grand bazaar. It holds a unique characteristic of traditional style of Persian architecture with related activities such Zoor Khaneh which was like an ancient gym in Iran.
Due to traffic zoning and land pricing upon economic crisis, the neighborhood was not quite vibrant nor safe place for people. The lots were too small and accessibility was interrupted. There were a lot of abandoned spaces around the neighborhood which made you feel unsafe.
We had a lot to do.









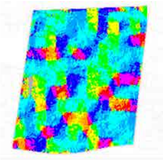


Site Analysis

After the document and field studies were conducted, we kicked off the site analysis. The geographical characteristics such as elevation levels, and aspect were analyzed by ArcMap tools before applying the network analysis of public transport stations across the neighborhood. At the end, we managed to perform the ground visibility and network integration analysis by IsoVist and Depthmap softwares.



The digital elevation model showed us the height levels of the site from sea level.



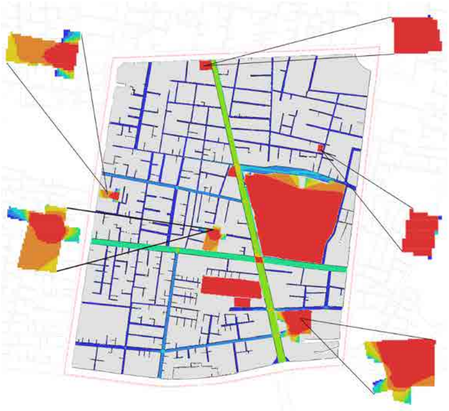
The aspect map helped us to realize the slope sides and the main southern aspect.



The street network analysis showed the access to bus stops within the 5 min walk

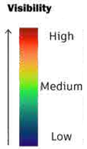


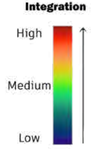
The street network analysis on metro stations presented the 1 km accessibility distance

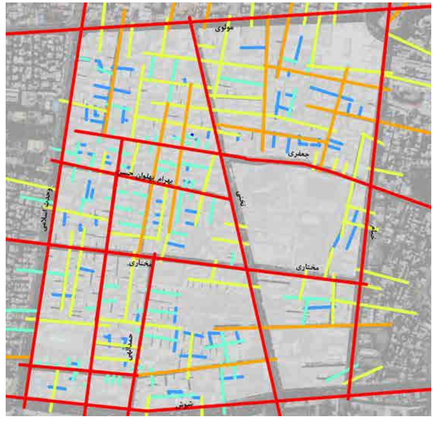


The visibility analysis was done with empy lots of the site to check wether if there's a proper sight across the field or not.

On the other hand, the street network integration analysis helped to identify the high integrated streets that are vital for activity flow.

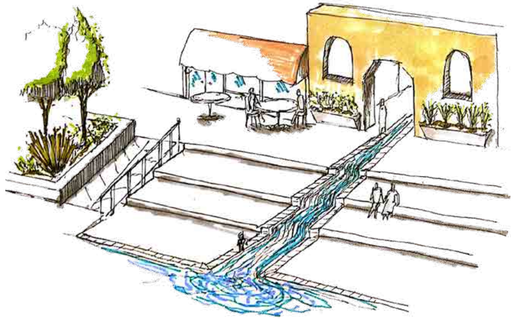


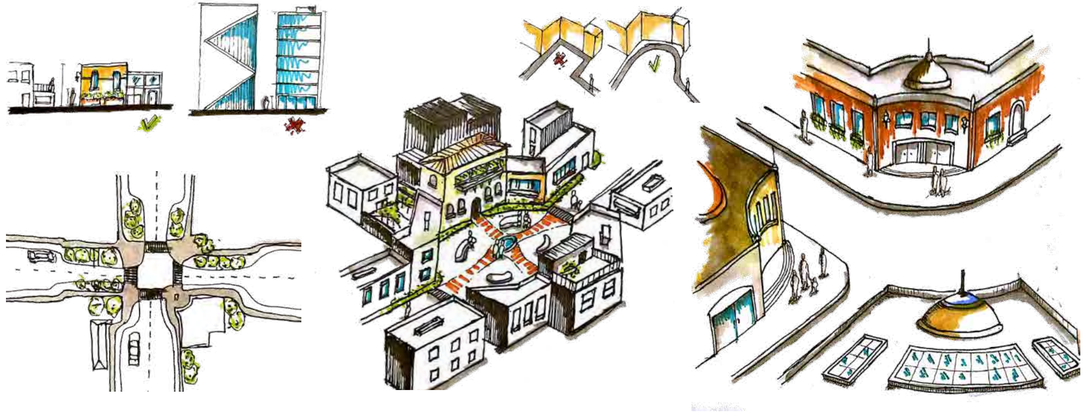




Design / Guidelines & Concept

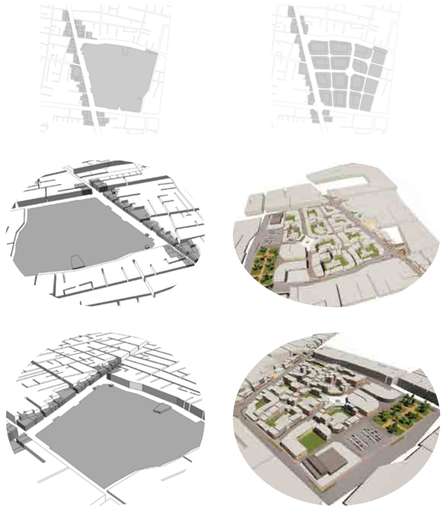
After identifying the local situations and contextual characteristics, we needed to create an urban design guideline based on traditional and historical forms of Persian architecture so that we could rebuild the old identity of the neighborhood, alongside the attempt to emerge the missing vibrancy and sense of place across the neighborhood spots.
Design guidelines were followed from standard considerations in urban design with a concentration on the social-cultural context of the design site.
As can be seen in hand-drawn figures, the guidelines are oriented around landscape, floor density, public space, streets and junctions design.

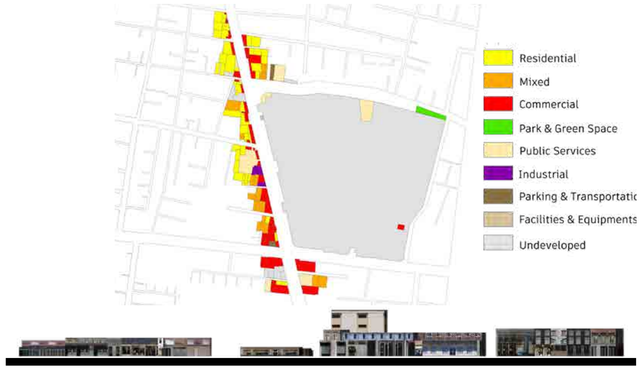




Planning & Design

After the guidelines were set and finalized, the project team decided to apply the mentioned guidelines on the present condition context. The vast empty lot inside the neighborhood was chosen as the design site. The process started with block design and land use planning simultaneously. The offset subdivision was the suitable block pattern because of its capability in holding mixed use with proper space in a single block. Due to the small lot & building patterns, this type helped me to develop enough space for all uses. Based on the activity needs and placement of diverse uses, the land use planning was set.
The 2D maps were created with AutoCAD & Arcmap.









The detailed designs were applied by SketchUp and on block scale and facade style, CityEngine helped to extrude the forms.
A multi-functional & interactive space with water and staircase elements were used. Using sharp stretched arcs for Persian style entrances.

Using tree lines on the subdivision entrances was considered alongside of setting corner buildings with moderate floor densities so the form can emphasize on the entrances.

At the crossing of three streets, the corner building was pushed back for extra open space.
Transparent facades and attractive activities with open corridors made by strip forms along the path.
Extension of sidewalks at the crossing were taken into account as a safety consideration.

Creating open local spaces among the blocks helps to emerge social interactions between the neighbors and residents.
Pushing back the buildings in squares helped to widen the open space.

